

CAPITAL CITY, KY, USA

WMO: 724233

Lat: 38.185N

Lon: 84.903W

Elev: 804

StdP: 14.27

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 53841

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 8.3	(c) 13.5	(d) -1.8	(e) 5.1	(f) 11.7	(g) 3.0	(h) 6.6	(i) 15.7	(j) 23.1	(k) 52.5	(l) 20.0	(m) 49.7	(n) 4.4	(o) 290	(p) 0.439

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 18.9	(c) 91.8	(d) 73.8	(e) 90.0	(f) 73.7	(g) 87.9	(h) 73.0	(i) 77.6	(j) 87.4	(k) 76.4	(l) 85.6	(m) 75.2	(n) 83.7	(o) 7.4	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.7	133.9	82.9	73.3	127.4	81.0	72.5	123.8	80.1	41.5	87.3	40.3	85.6	39.1	83.9	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 18.9	(b) 16.7	(c) 14.7	(e) 0.1	(f) 95.1	(g) 7.0	(h) 3.6	(i) -4.9	(j) 97.7	(k) -9.0	(l) 99.8	(m) -12.9	(n) 101.9	(o) -17.9	(p) 104.5		
(4) 18.9	(5) 16.7	(6) 14.7	(7) 0.1	(8) 95.1	(9) 7.0	(10) 3.6	(11) -4.9	(12) 97.7	(13) -9.0	(14) 99.8	(15) -12.9	(16) 101.9	(17) -17.9	(18) 104.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	56.0	33.4	36.9	45.2	56.4	65.1	72.8	76.1	75.3	69.0	57.0	45.6	37.5
	DBStd	17.50	12.15	11.25	10.61	9.09	7.67	5.55	4.59	4.72	7.06	8.96	9.33	10.57
	HDD50	1817	531	386	221	41	3	0	0	0	34	192	409	
	HDD65	4523	978	786	616	280	98	10	1	40	279	582	852	
	CDD50	3997	18	20	74	234	471	683	810	785	570	250	61	21
	CDD65	1229	0	1	3	23	101	243	346	321	160	30	1	0
	CDH74	11186	0	2	31	209	873	2171	3162	2881	1553	295	9	0
	CDH80	3993	0	0	2	23	215	750	1234	1116	585	67	1	0
(14) Wind	WSAvg	6.0	7.4	7.1	7.3	7.4	5.9	5.3	4.6	4.4	4.5	5.3	6.1	6.7
(15) Precipitation	PrecAvg	44.8	3.1	3.1	4.4	3.9	4.8	4.4	4.4	3.6	3.4	2.7	3.4	3.6
	PrecMax	58.5	6.1	8.3	14.3	8.4	11.3	10.1	8.5	8.7	12.1	7.1	8.2	9.8
	PrecMin	32.3	0.4	0.6	1.2	0.8	1.3	0.5	0.4	0.4	0.5	0.1	0.4	0.5
	PrecStd	7.1	1.6	1.7	2.7	2.0	2.2	2.0	2.1	1.7	2.4	1.5	1.6	1.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.8	70.9	78.7	82.7	87.8	91.6	95.2	95.4	93.7	85.6	74.7	67.7
		MCWB	59.2	60.0	63.0	64.0	71.5	73.4	77.0	74.2	71.0	68.5	62.6	60.0
	2%	DB	62.5	64.3	72.7	79.6	84.6	89.5	91.3	91.4	90.3	81.3	70.0	62.8
		MCWB	57.9	55.3	59.4	63.7	69.2	73.6	75.4	73.4	71.3	67.0	59.3	57.0
	5%	DB	57.6	60.2	67.7	76.0	82.3	87.5	89.6	89.0	87.0	76.9	65.5	58.2
		MCWB	53.6	53.3	56.5	61.7	68.5	72.7	75.0	73.5	70.4	64.1	57.6	53.5
	10%	DB	53.4	55.0	63.2	72.5	79.6	84.5	87.0	86.3	82.6	72.5	61.5	54.2
		MCWB	48.3	48.3	54.2	60.1	67.5	71.4	73.9	72.5	68.7	62.6	54.5	50.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.6	61.7	64.5	68.4	74.3	77.7	80.3	78.5	76.0	73.3	65.2	61.9
		MCDB	63.8	69.0	75.4	79.1	83.3	87.3	88.8	88.4	86.4	83.1	72.3	66.6
	2%	WB	58.0	58.2	61.6	66.2	72.1	76.2	78.1	77.1	74.2	69.6	62.2	58.1
		MCDB	61.5	63.2	69.8	75.8	81.5	85.4	87.8	86.6	84.4	76.4	67.5	61.6
	5%	WB	54.5	54.2	59.0	64.0	70.5	74.8	76.9	75.6	72.6	66.9	58.8	54.3
		MCDB	57.7	58.8	66.0	72.1	79.4	83.7	86.4	84.4	81.9	74.2	63.9	57.3
	10%	WB	48.7	48.8	55.6	61.9	68.7	73.4	75.5	74.4	71.2	64.1	55.2	50.2
		MCDB	52.8	54.7	62.4	69.5	76.7	81.9	83.6	82.5	79.4	70.4	60.6	54.0
(35) Mean Daily Temperature Range	5% DB	MDBR	16.4	18.0	20.2	21.5	20.3	19.7	18.9	20.0	22.1	21.5	19.8	15.8
		MCDBR	19.2	24.0	25.3	25.5	23.2	22.6	21.7	23.3	26.9	26.0	24.7	20.2
		MCWBR	15.7	17.9	15.3	13.8	10.9	10.0	8.4	8.8	10.6	13.2	16.3	16.4
	5% WB	MCDBR	18.0	21.5	21.1	20.6	19.5	18.9	18.5	18.2	20.3	20.7	20.2	18.1
		MCWBR	15.6	18.2	14.0	13.2	10.4	9.6	8.6	8.3	9.6	12.0	15.8	16.1
(40) Clear-Sky Solar Irradiance	taub		0.310	0.324	0.353	0.394	0.438	0.451	0.474	0.468	0.413	0.356	0.324	0.302
	taud		2.464	2.403	2.387	2.269	2.192	2.193	2.136	2.145	2.271	2.430	2.476	2.490
	Ebn at Noon		276	286	288	280	269	265	257	256	264	271	269	271
	Edh at Noon		26	31	35	42	46	46	49	47	39	30	25	23
(44) All-Sky Solar Radiation	RadAvg		607	827	1172	1561	1747	1962	1857	1742	1437	1068	735	523
	RadStd		65	117	108	114	161	104	107	99	147	118	82	52

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+227	+139

Nomenclature: See separate page